

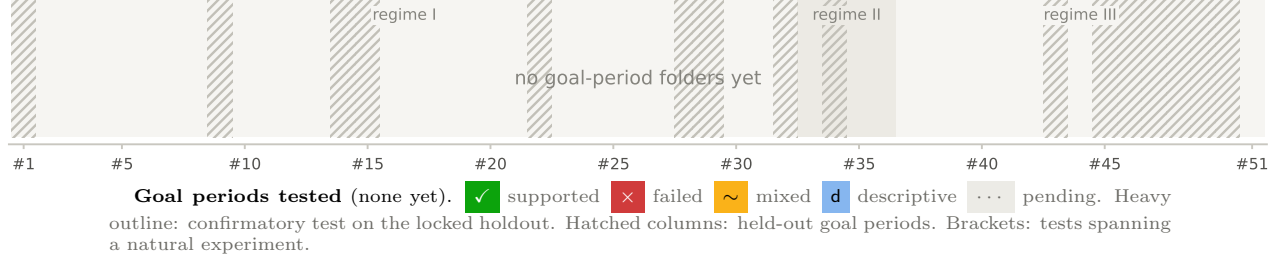
# H09: Agent swarms have an effective thermodynamics

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 25% Faithfulness 1.5/5 Usefulness 2/5

**Bottom line.** As a framing, swarm thermodynamics has not yet earned its keep; its best result is a practical fact about scheduling: in the current scaffold, idle agents are timer-gated, and only @-mentions get through.

level: hypothesis · pipeline checks 14% · updated 2026-10-04



## WHAT HAPPENED

An exploratory framing, not one test: can swarm statistics be read as thermodynamics (landscapes, currents, traps, equations of state)? Two light non-holdout rounds; no score-card or holdout run. Round 1: the activity landscape's two wells are reproduced by the daily schedule (a field, not coupling). Per-agent action chains are irreversible (78% of cells beat the null), with a closed regime-III cycle consolidate → search → talk → pause, but event-level entropy production collapses after perma-computer-use. Agents are ergodic at day scale; restarts relax within a minute. Round 2 built reusable tables (`idle_runs`, `consolidation_inflow`). In regime I, room messages trigger escape from idling (hazard ratio 3.06; null p95 0.96). In regime III a pause is a self-set timer (0.22% end early), and only @-mentions matter ( $\times 1.52$ ), at expiry. Memory size is an agent-specific set point, reset at each consolidation and barely moved by inflow. Collective entropy production: small (plug-in only).

## PHYSICAL MODEL

No single model; each sub-claim borrows one (models 01, 02, 09, 10).  $\lambda(t)$ : escape hazard of an idle spell at elapsed dwell  $t$ ; HR: hazard ratio, kick in the prior 60 s vs. none.  $k_0$ ,  $c$ : base escape rate and kick sensitivity;  $r$ : kick rate.  $t^*$ : declared pause expiry;  $p_{\text{gate}}$ : chance of acting at expiry.  $\pi$ ,  $P$ : stationary law and transitions of an agent's action-class chain.  $V$ : memory size;  $P_{\text{in}}$ : inflow;  $\beta$ ,  $\gamma$ : response, relaxation.

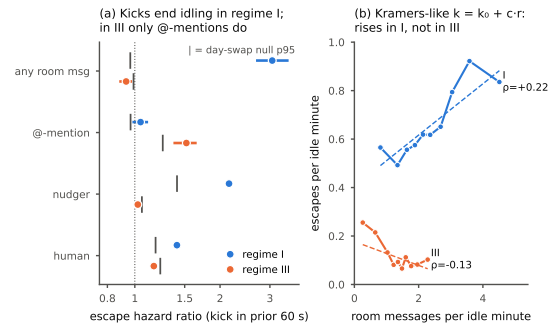
$$\lambda(t) = P(\text{escape at } t \mid \text{idle until } t), \quad \text{HR} = \lambda_{\text{kick}} / \lambda_{\text{none}}$$

regime I (Kramers-like):  $k \approx k_0 + cr$

regime III (timer gate):  $\lambda(t) \approx \delta(t - t^*) p_{\text{gate}}(\text{@-mention})$

$$\sigma = \sum_{x < y} (\pi_x P_{xy} - \pi_y P_{yx}) \ln \frac{\pi_x P_{xy}}{\pi_y P_{yx}}$$

$$\Delta V = \beta P_{\text{in}} - \gamma V_{\text{prev}} \Rightarrow V^* = \beta P_{\text{in}} / \gamma$$



**Observables.** (a) Escape hazard ratio of idle spells: minutes with a kick in the previous 60 s vs. minutes without, stratified by elapsed dwell and time of day. Bars: agent-day bootstrap CIs where computed; ticks: 95th percentile of a day-swap null. Regime I: any room message triples escape. Regime III: room messages don't help; only @-mentions do. (b) Escape rate against kick rate over agent-day deciles (dashed: linear fit  $k_0 + cr$ ). It rises in I and falls in III.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The thermodynamic vocabulary did little work; the useful findings are plain mechanics. The one an operator can use: in the current scaffold (regime III), an idle agent is asleep on a timer it set itself. Room chatter does not wake it. Only an @-mention raises the chance it acts, and only when the timer expires. To get a response, address the agent by name and expect a delay of up to its chosen pause. Regime-I agents were interruptible. "Memory size is a set point" says consolidation is a reset, not growing storage. The rest restates the scaffold: landscapes made by the schedule, day-scale ergodicity, and a consolidation cycle that is the prompt's own loop. Nothing here yet beats a simpler statistic.

H09-swarm-thermodynamics · generated 2026-10-03 18:07 by in-fra/summaries/build\_summaries.py · card hypotheses/H09-swarm-thermodynamics/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))